WinUI 3 Fundamentals

Creating Your First WinUI Application

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Version Check

Version Check

This version was created by using:

- WinUI 3 / Windows App SDK 1.0
- .NET 6
- Visual Studio 2022

Version Check

This course is 100% applicable to:

- WinUI 3 / Windows App SDK 1.0
- .NET 6
- Visual Studio 2022

Module Outline

What is the Windows UI library (WinUI)?

What you will learn in this course

Create your first WinUI app

- Set up your environment
- Create and explore a WinUI project
- Understand how files are generated

What Is the Windows UI Library (WinUI)?

Native UI framework of Windows 10/11

Build Windows desktop apps with C# or C++

WinUI ships as part of the Windows App SDK

- Project templates
- NuGet packages

WinUI is open source

https://github.com/microsoft/microsoft-ui-xaml

WinUI Features for Application Development

What you need to know to build and maintain WinUI apps

XAML
(Extensible Application Markup Language)

Data binding
MVVM

Layout

Styles and Templates

https://docs.microsoft.com/windows/apps/desktop

What You Will Learn in This Course

No prior WinUI knowledge is required

Create and maintain WinUI apps

- XAML
- Layout
- Data binding
- MVVM
- Resources
- Styles & Templates

You will work with me for the Wired Brain Coffee company

Course Structure

- Creating Your First WinUI Application
- Instantiating Objects in XAML
- Building a User Interface
- Organizing Code with UserControls
- Applying Data Binding and MVVM
- Executing Code with Commands
- Creating Reusable Resources
- Working with Data Templates
- Setting up Dependency Injection
- Styling Your Application
- Mastering Control Templates
- Debugging Your WinUI Application

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Download exercise files from the course homepage on pluralsight.com

The exercise files contain the code before and after every module

Set up Your Environment

Install Visual Studio 2022

Community edition

https://www.visualstudio.com

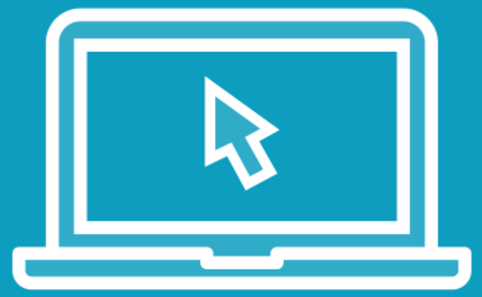
Select the required workloads

.NET desktop development

Windows App SDK C# templates

Universal Windows Platform development

Demo



Check your Visual Studio installation

Find installation instructions in the Microsoft documentation

Demo

Create and explore a WinUI project

Demo

Work with XAML and code-behind

Demo

Understand how files are generated

Summary

WinUI is the native UI framework of Windows 10/11

- Build desktop apps for Windows

Define user interfaces with XAML

The main window consists of multiple files

- MainWindow.xaml
- MainWindow.xaml.cs (code-behind)
- More files are generated in obj\Debug

Up Next:
Instantiating Objects in XAML